Massachusetts Institute of Technology

Design Review ~Drivetrain~

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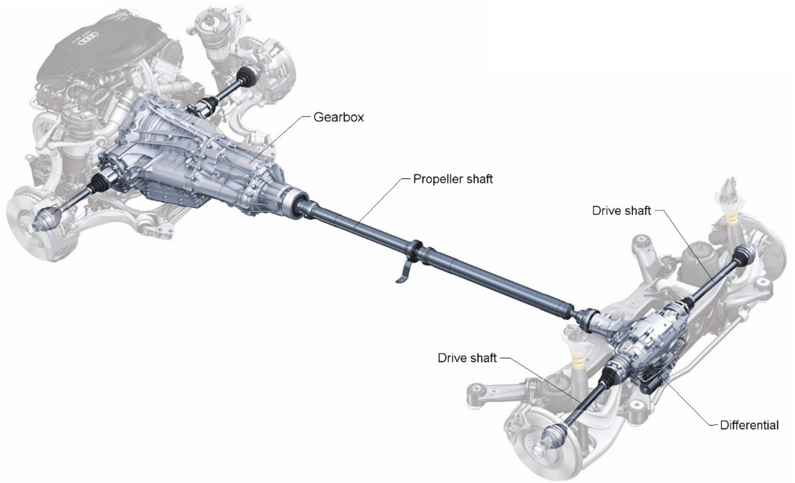
Sub Assembly Team Overview

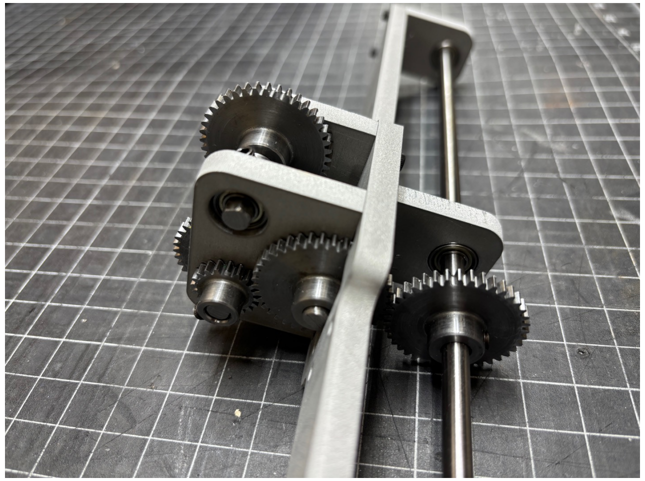
What is drivetrain? Group of components that deliver mechanical power to the wheels.

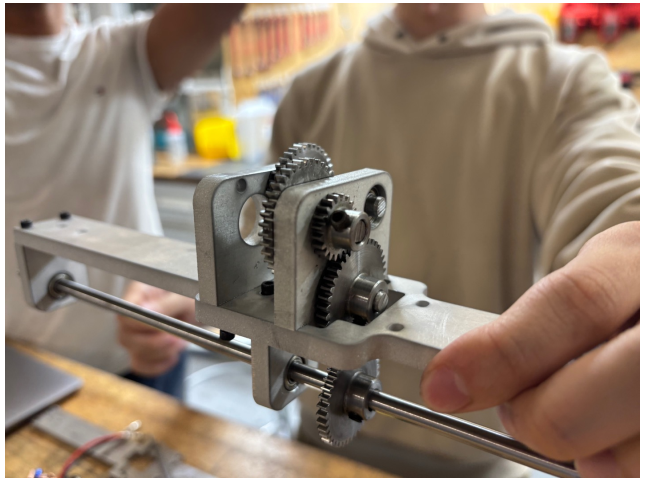
Current Status:
- CAD phase with various designs being considered
- 3D printed prototypes and waterjet assembly made
- Preliminary calculations on required gear ratios and torque



Challenges Addressed:
- Reducing speed (issue from last year)
- Optimal gear ratio and alignment

Future Challenges:
- Suspension integration
- FEA simulation/testing
- Ease of manufacturing (mass production)
- Tolerancing issues





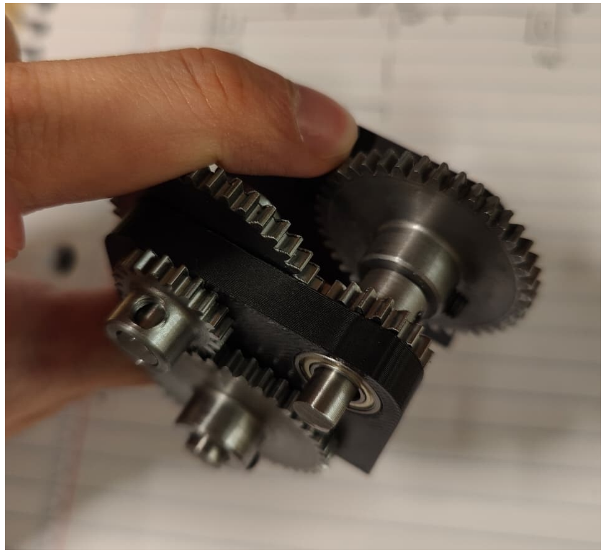
Design Considerations

Gear Ratio Analysis & Design Decisions

Objective: Based on motor specifications, achieve an adequate torque multiplication for navigating obstacles and ramp while maintaining a reasonable wheel speed.

Motor Constraints

- **Motor:** Reddy Radon 2, 17T, 3600Kv
- **Input Voltage:** 7.4 V LiPo
- **Motor RPM:** 26,600 RPM (3600 x 7.4V)
- Motor shaft diameter limits it to a 12T pinion gear

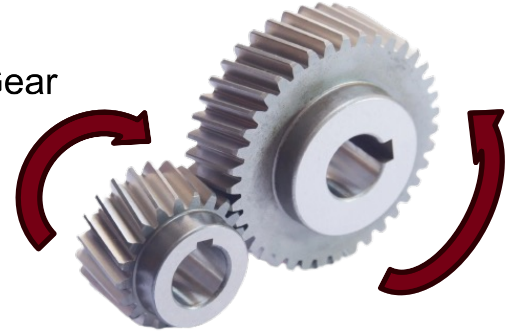


Note: RPM & Speed Calculations assume theoretical maximum (friction, car mass, gravity will slow the speed down by ~60-70%)

Gear Reduction Design

- **13.33x Reduction:** 12 → 40, 20 → 40, 20 → 40
- **Wheel RPM (no-load):** ~1998 RPM
- **Motor RPM:** 26,600 RPM (3600 x 7.4)
- **No-Load Linear Speed (Assume D = 6 in):** 15.95 m/s

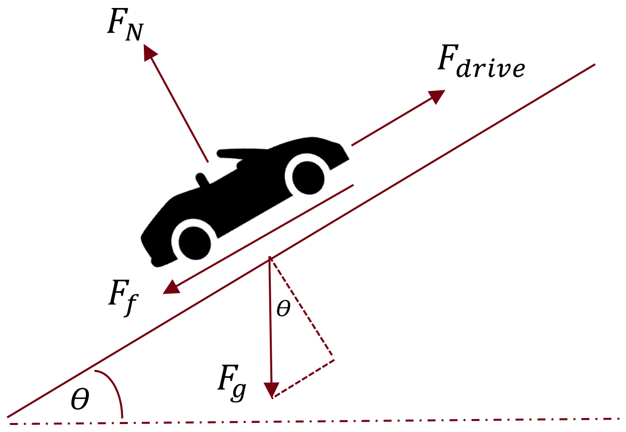
↓ Speed $Gear\ Reduction = \frac{Driven}{Driving}$ ↑ Torque



Design Considerations

Ramp Climbing Feasibility Analysis

Objective: Determine whether the motor provides sufficient torque for the vehicle to climb the ramp based on certain assumptions.



$$F_{drive} = mgsin\theta + \mu mgcos\theta$$

$$T_{axle,req} = \frac{F_{drive} r_{wheel}}{Gear\ Reduction}$$

$$T_{motor} = \frac{P}{\omega} = \frac{21.6W}{(26640\ RPM\ * \frac{2\pi}{60})} = 0.00774\ Nm$$

$$T_{axle,available} = T_{motor} * Gear\ Reduction$$

Assumptions

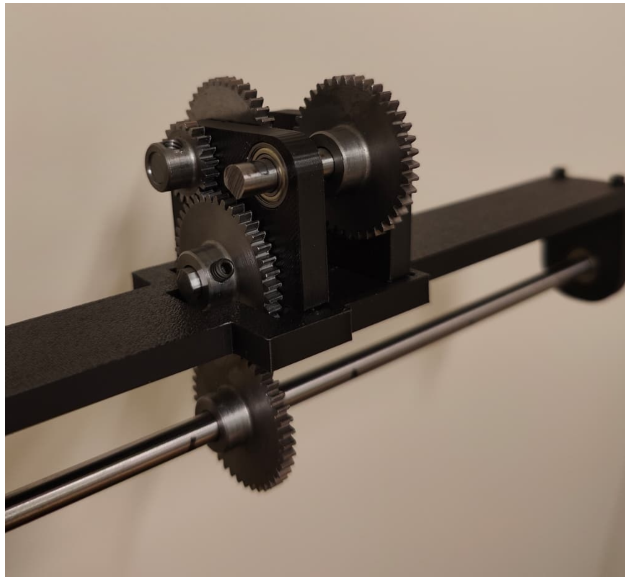
- Wheel diameter (D = 6”)
- Gravity (g = 9.81 m/s^2)
- RC car mass cases: 3 lbs., 5 lbs.
- Friction coefficient: 0.1
- Maximum Ramp Angle: 45°
- Max Battery Power: 21.6 W

Mass	$F_{gx} + F_f$ (N)	$T_{axle,req}$ (Nm)	$T_{axle,avail}$ (Nm)
3 lbs. (~1.36 kg)	10.35	0.059	0.103
5 lbs. (~2.27 kg)	17.28	0.099	0.103

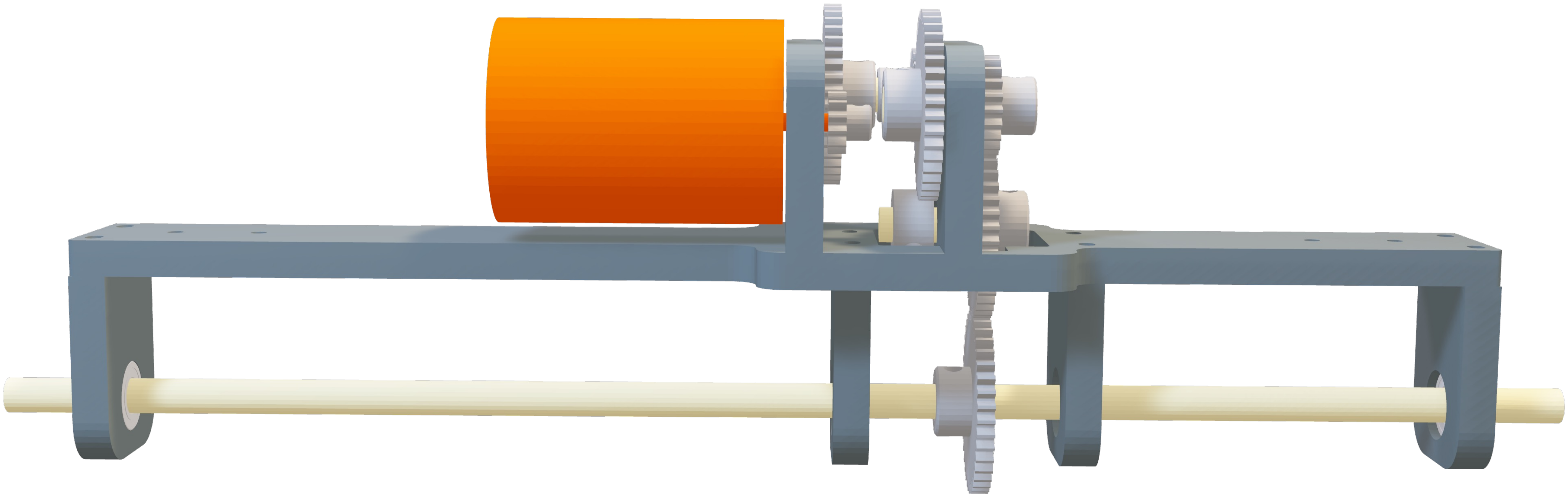
Design Considerations

Engineering Constraints → Design Requirements

Engineering Constraints	Design Requirements
Motor shaft only fits on 12 tooth gear due to shaft diameter	Pinion gear must be 12 tooth to mate with motor
Axle shaft diameter is 0.25” → spur gears must be 20 teeth or 40 teeth	Gear reduction combinations are limited to 12:40 and 20:40
Maximum ramp width is 16”, Minimum ramp width is 5.25”	Axle length must be within 5.25” and 16”
Vehicle must climb a maximum of 45° ramp	Torque provided must be greater than or equal to required torque to overcome friction + gravity
Vehicle Weight	Drivetrain must support roughly 3-5 lbs. of vehicle weight
Material Selection based on Manufacturability, Cost, & Available Stock	6061 Aluminum (0.25” Thick) for sufficient strength to support motor torque while being compatible with waterjet



CAD
Full Assembly



CAD
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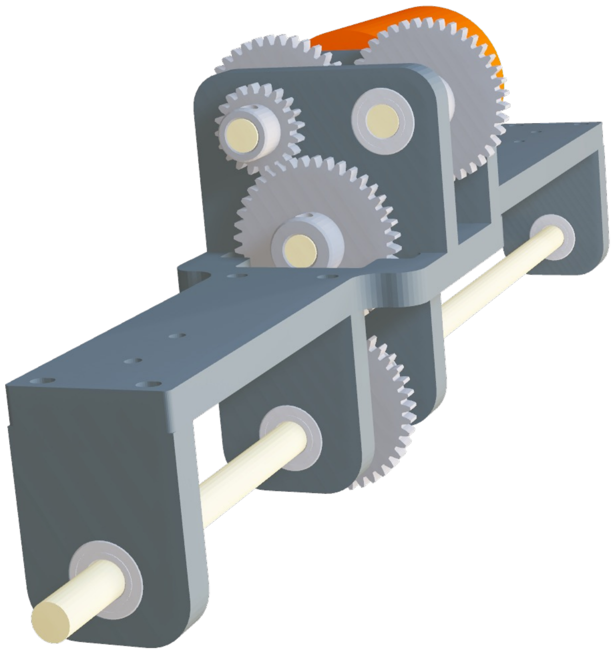
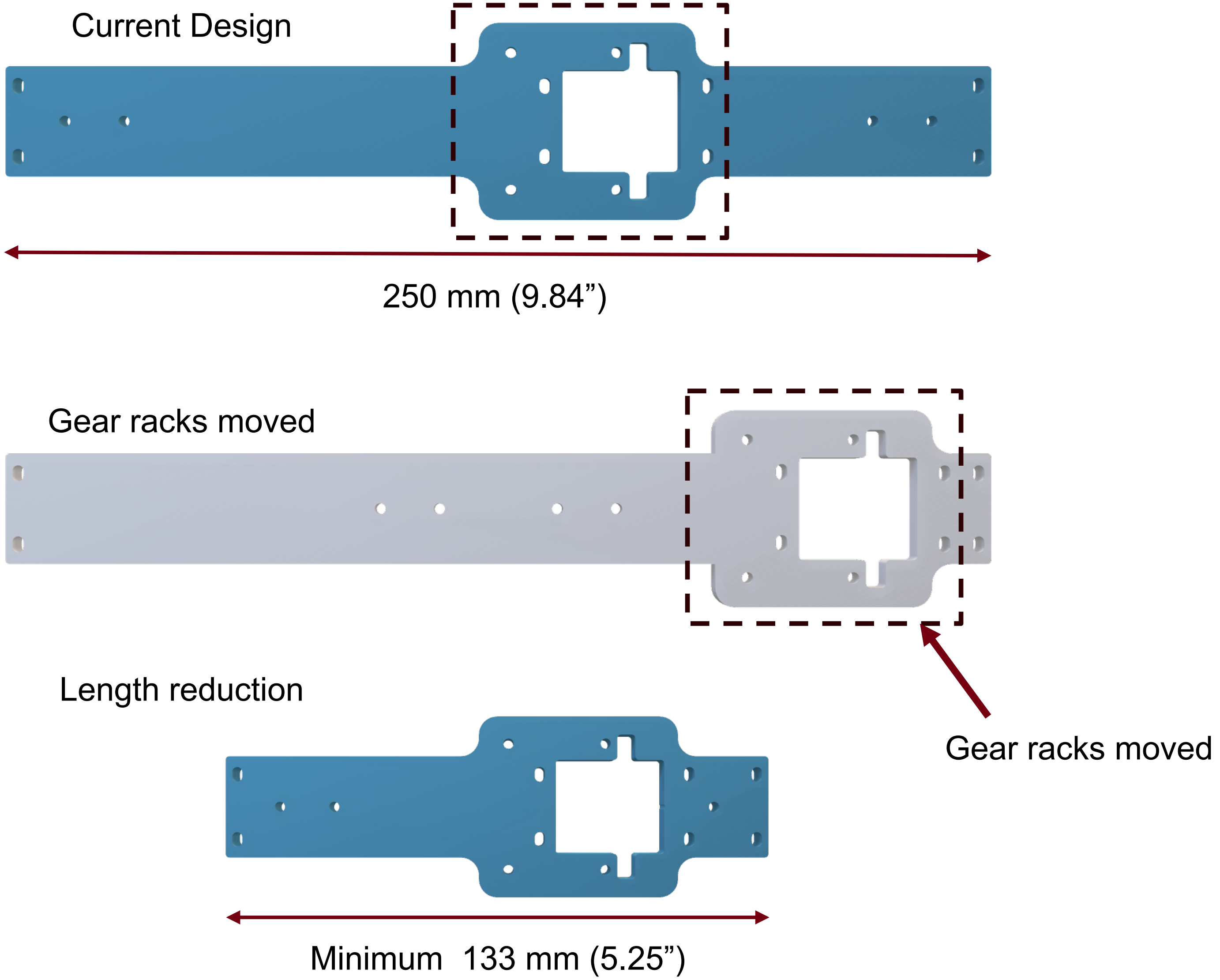
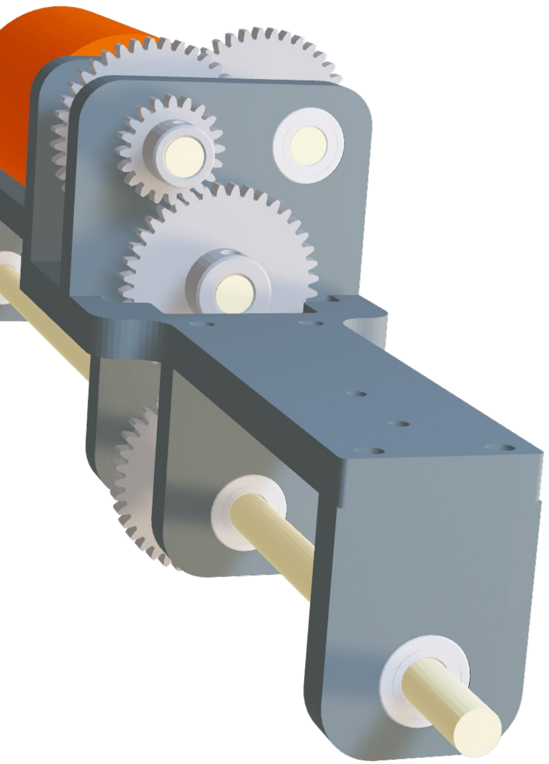


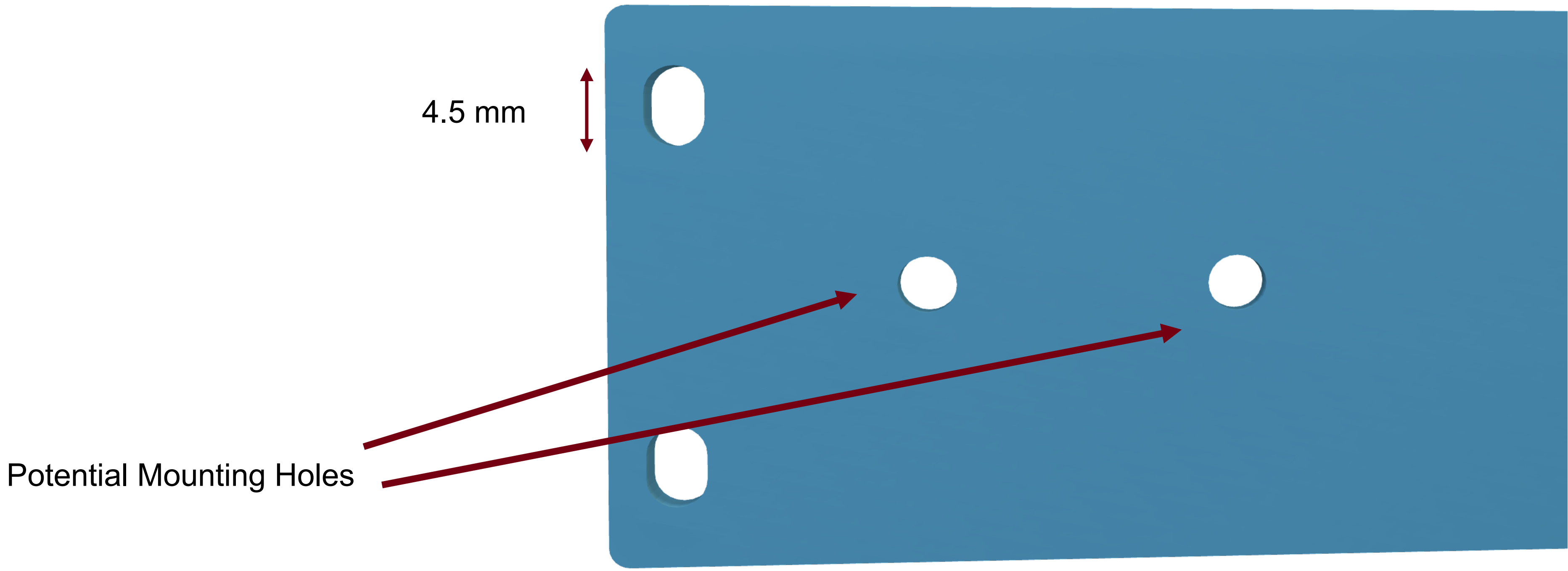
Design Considerations:
Movable Gear racks to allow space for length reduction or future mounting holes.
Axle length must be within 5.25” and 16”



CAD
Full Assembly

Design Considerations:
Enlarged axle holder mounting holes to account for variation.

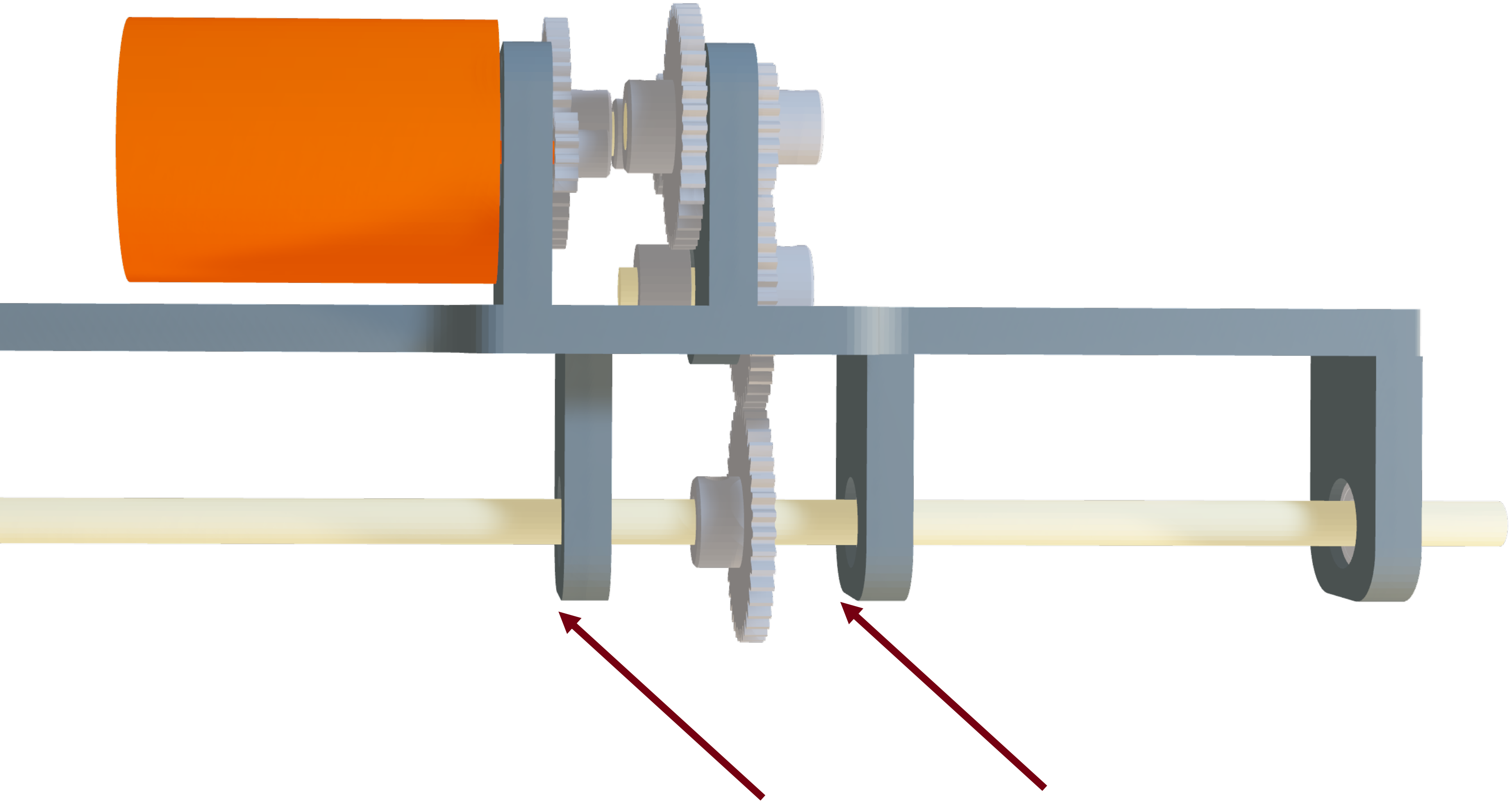




CAD
Full Assembly

Design Considerations:

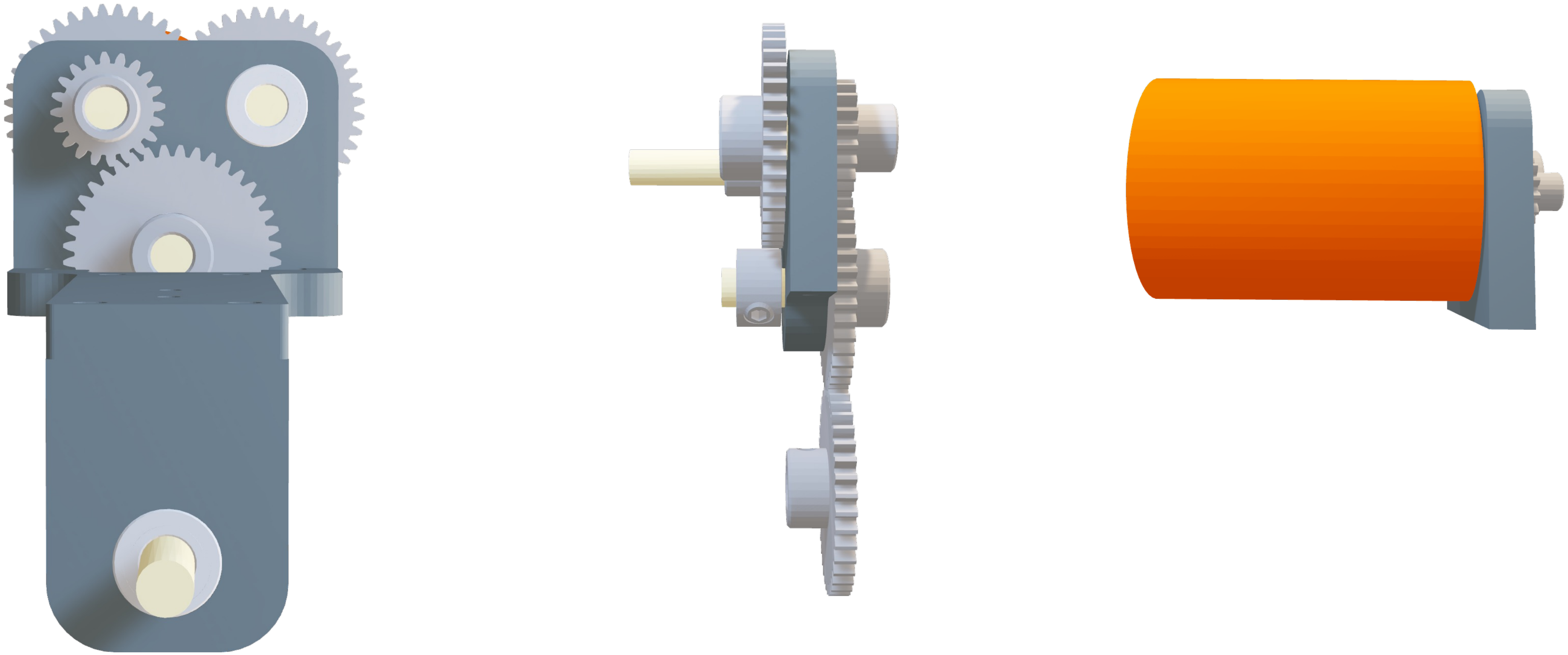
Axle holders designed to be identical to prevent rod deflections and allow for simple manufacture



Prevent rod deflections

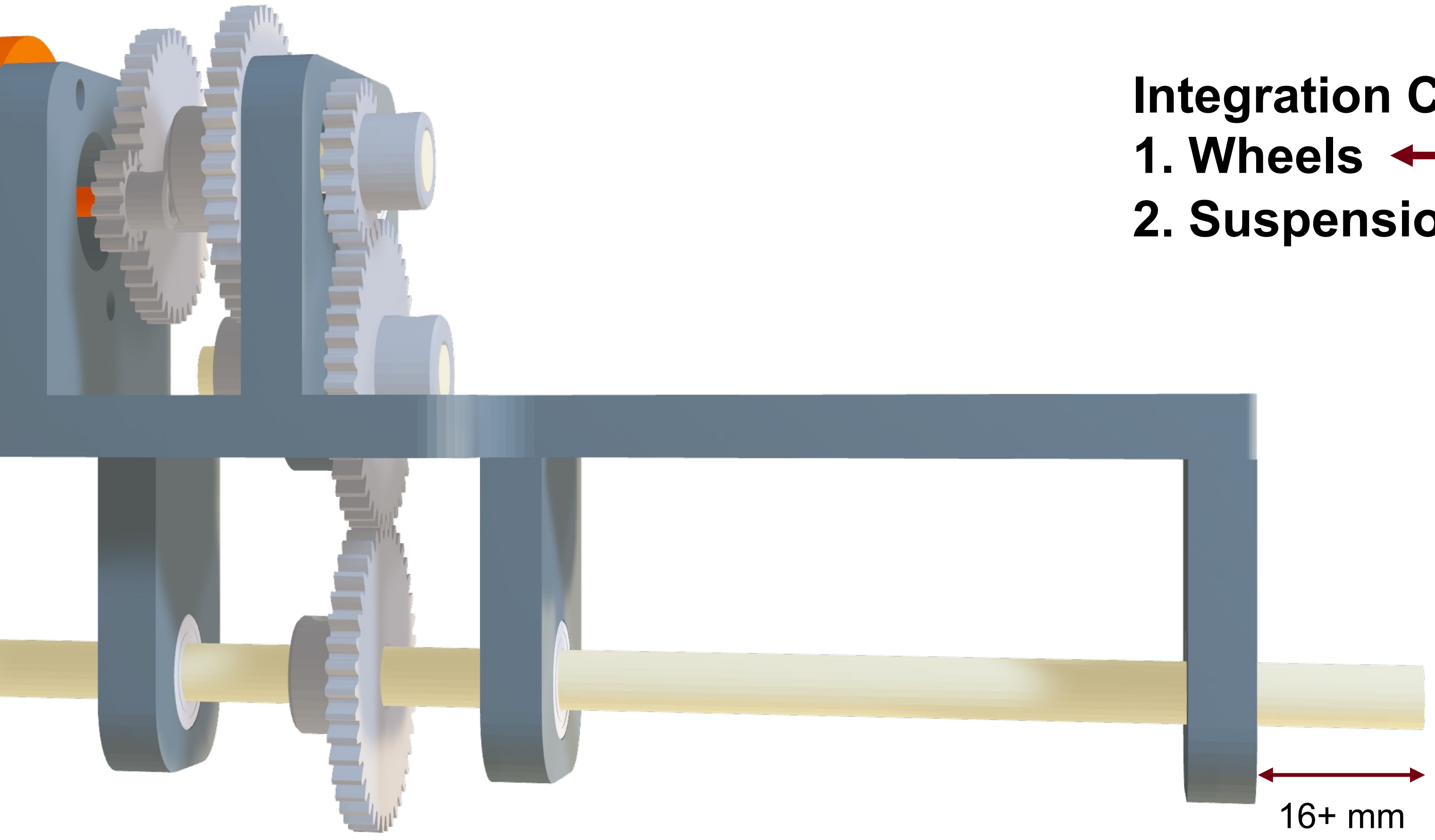
CAD
Full Assembly

Design Considerations:
Previous car gear reduction: **1:11.08**
This year's gear ratio : **1:13.32**
Goals:
- Slower speed than previous year
- Motor mount as low as possible



CAD

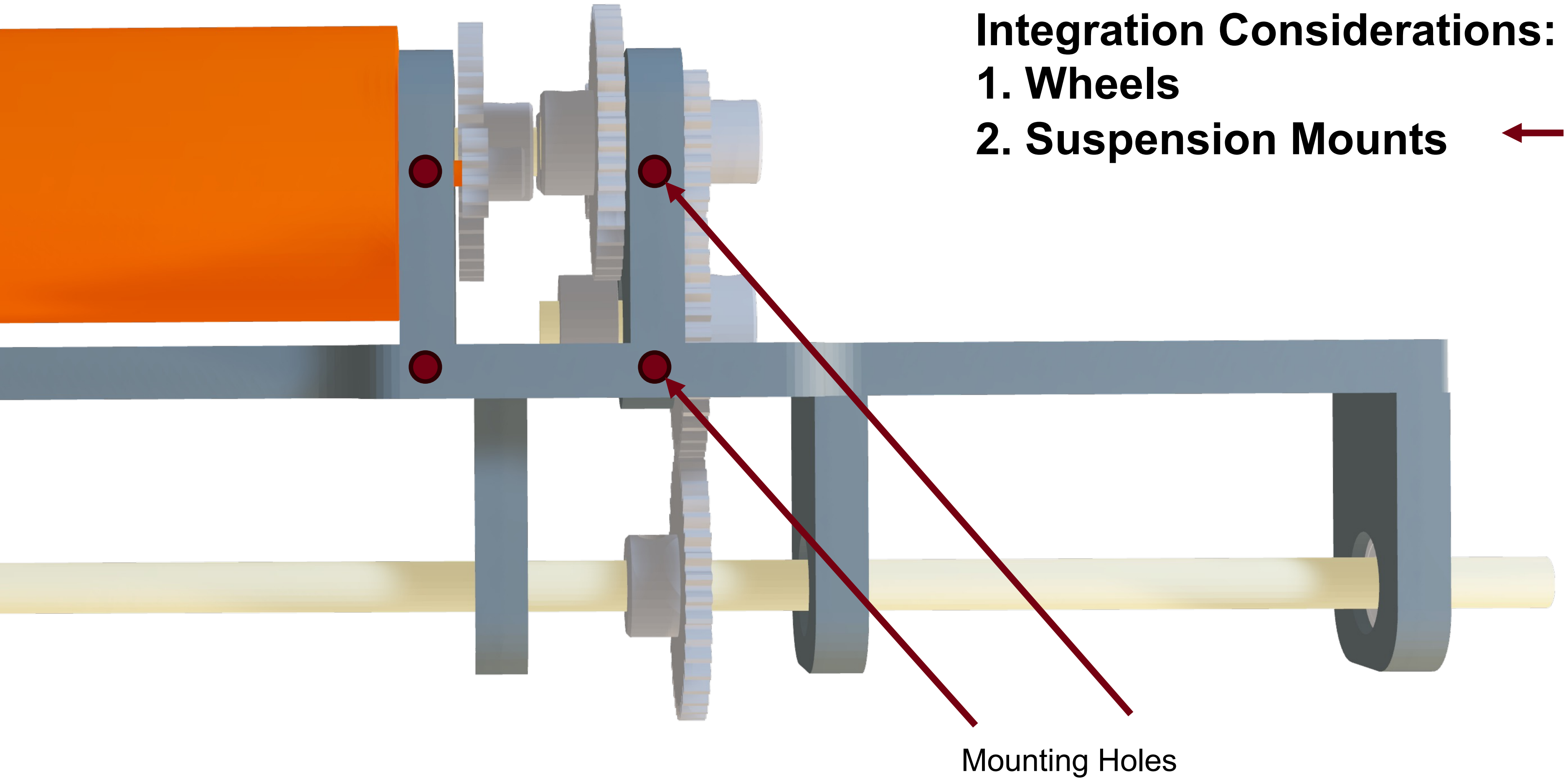
Full Assembly Integration



Integration Considerations:

1. **Wheels**
2. **Suspension Mounts**

CAD
Full Assembly Integration



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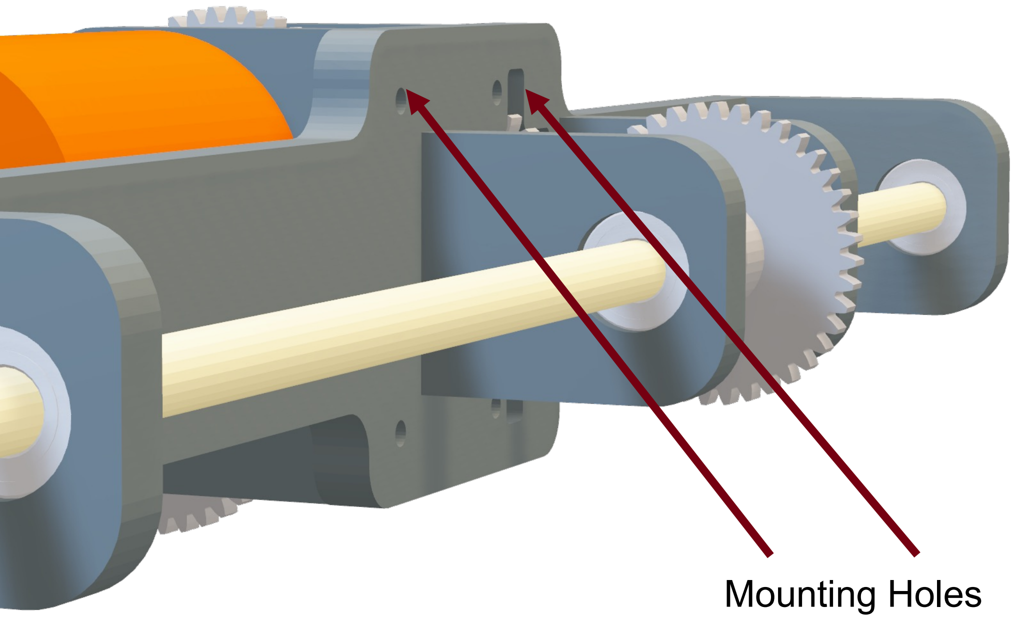
Full Assembly Integration



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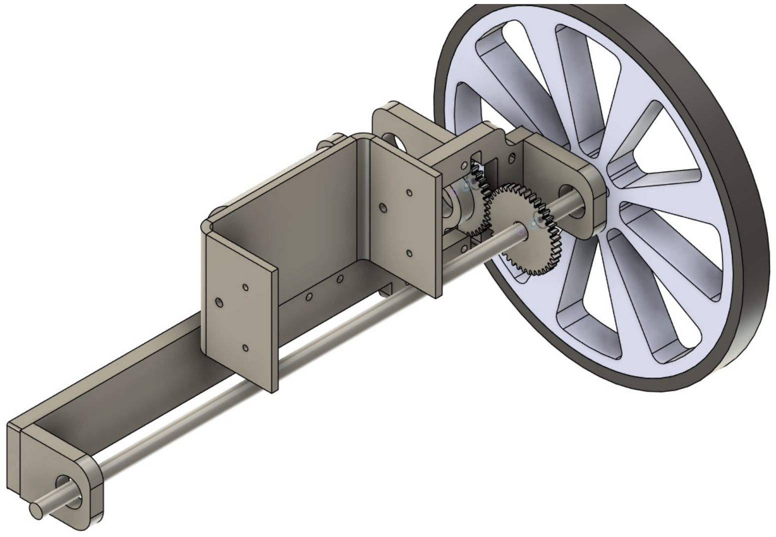
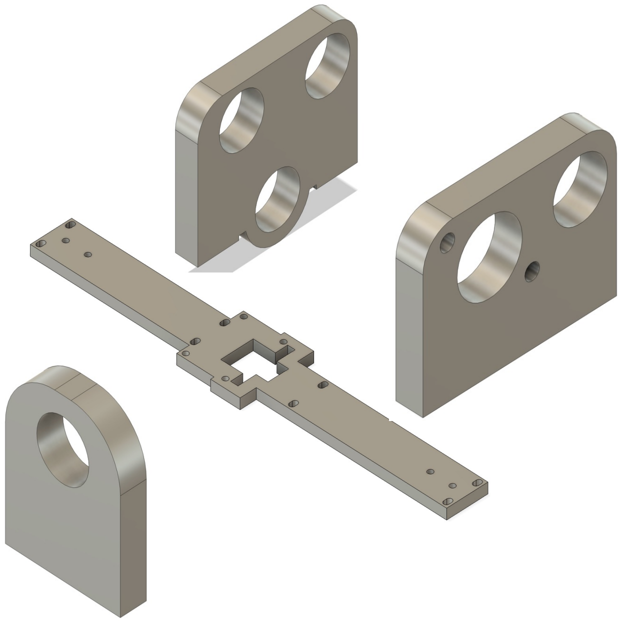
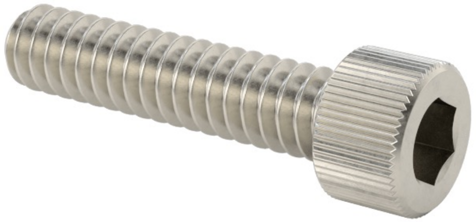


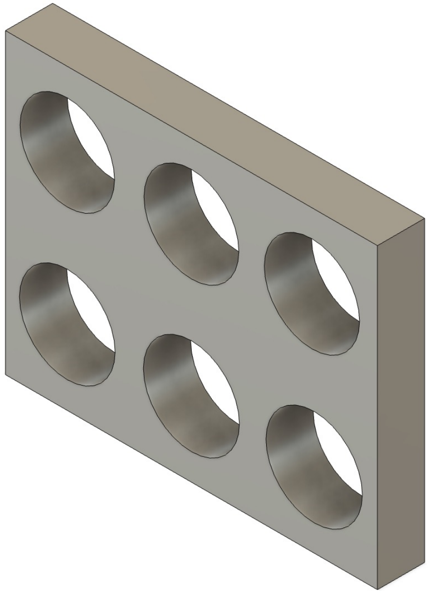
DFM/DFA Considerations



Manufactured components are of uniform thickness.

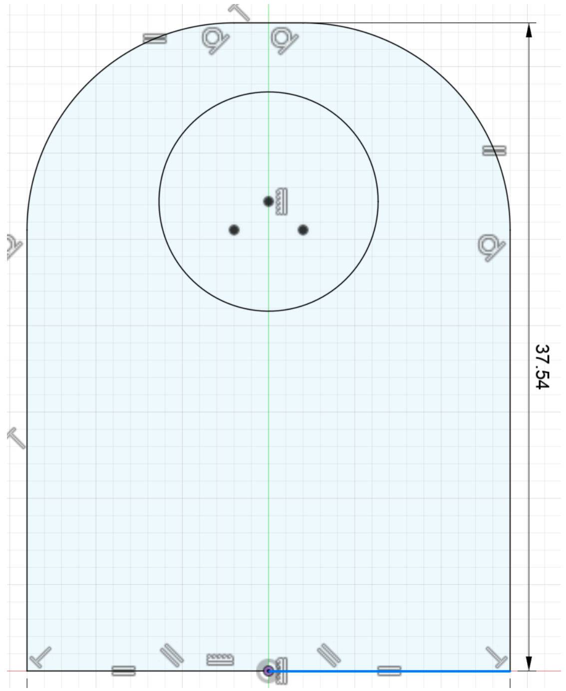


4-40 x ½ screws used throughout assembly.

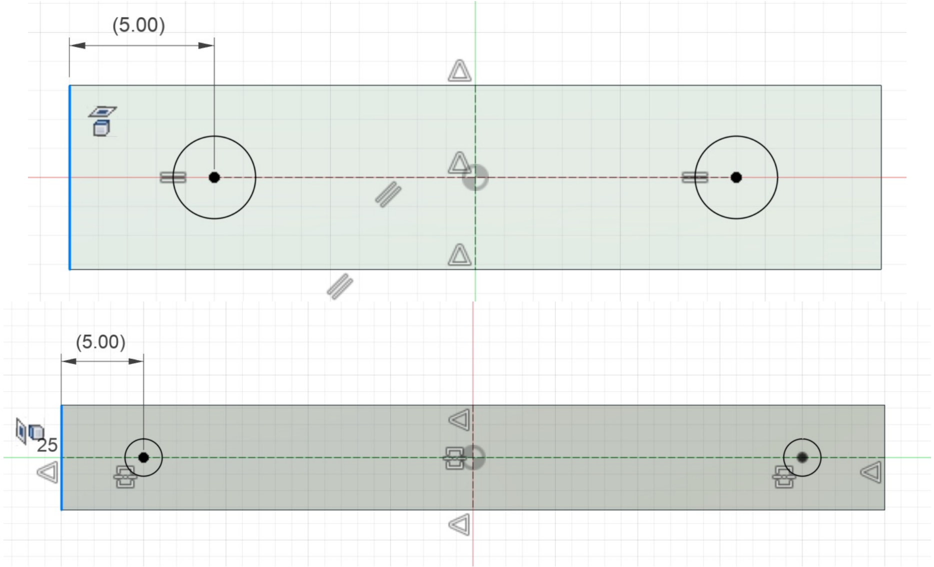


Bearing slightly oversized to remove extra reaming operations.

DFM/DFA Considerations



Axle holders, gear block, and motor mount are slightly elongated to allow for a machined, flat surface for mating.



Drilled holes are placed in the same location for both axle holders and gear block/motor mount.

Process Plan Example

2 extra reaming tasks if DFM not followed.

#	Task	Machine	Tooling
1	Cut part from stock	Water Jet	
2	Fixture part for machining	Mill Vise	Machining jig
3	Machine off 1mm from bottom	Mill	0.5” end mill
4	Fixture part for drilling	Mill Vise	Drilling jig
5	Spot drill holes	Mill	Spot drill
6	Drill holes	Mill	#43 drill bit 0.5” depth
7	Fixture part for tapping	Vise	
8	Tap drilled holes	Powered tap	4-40 tap

Applicable for Axle Holder, Motor Mount, and Gear Block.

Mfg. Equipment & Stock Procurement

Equipment Required

- Water jet – To cut flat, metal components
- Belt sander – Remove nubs from water jet parts and rough edges from cut axle shafts
- Bridgeport – Machine flat mating surface on certain components, drill mounting holes
  - Jigs used to decrease setup time
- Powered tap – Quickly tap mounting holes
- Vertical band saw – To cut axle shafts to length
- Drill press – Ream out all gears to allow for easier fitment on shafts

Stock Required

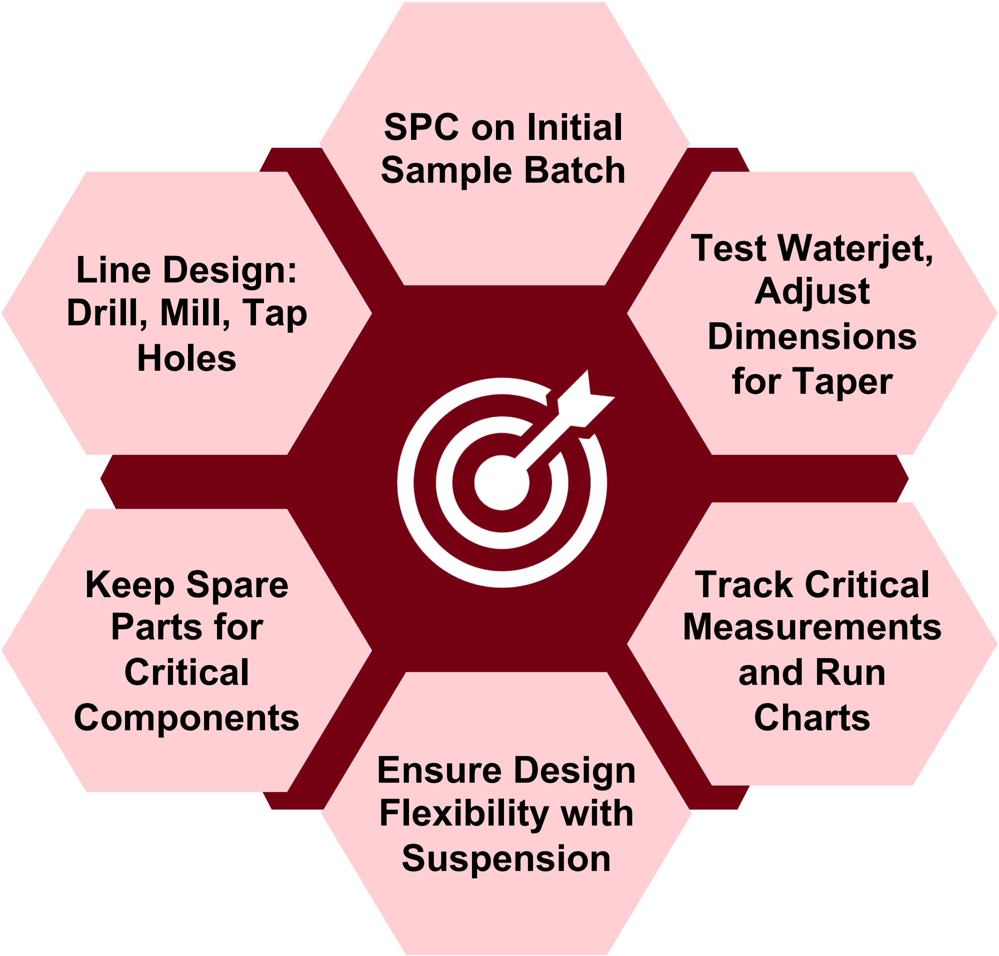
- Hardware (screws, set screws, bearings, shaft collars)
  - Ordered as of 10/11 - $532.32
- 6061 1/4 in. aluminum flat stock
  - Required area ~ 1080 in^2
  - Max stock size for water jet – 2 ft x 4 ft = 1152 in^2
  - Total cost of stock from McMaster - $247.47
- Using course-provided stock and hardware for testing purposes

Strategic Plan for Drivetrain Fabrication

Objective: Efficiently fabricate 40 drivetrain assemblies on time while ensuring assembly quality

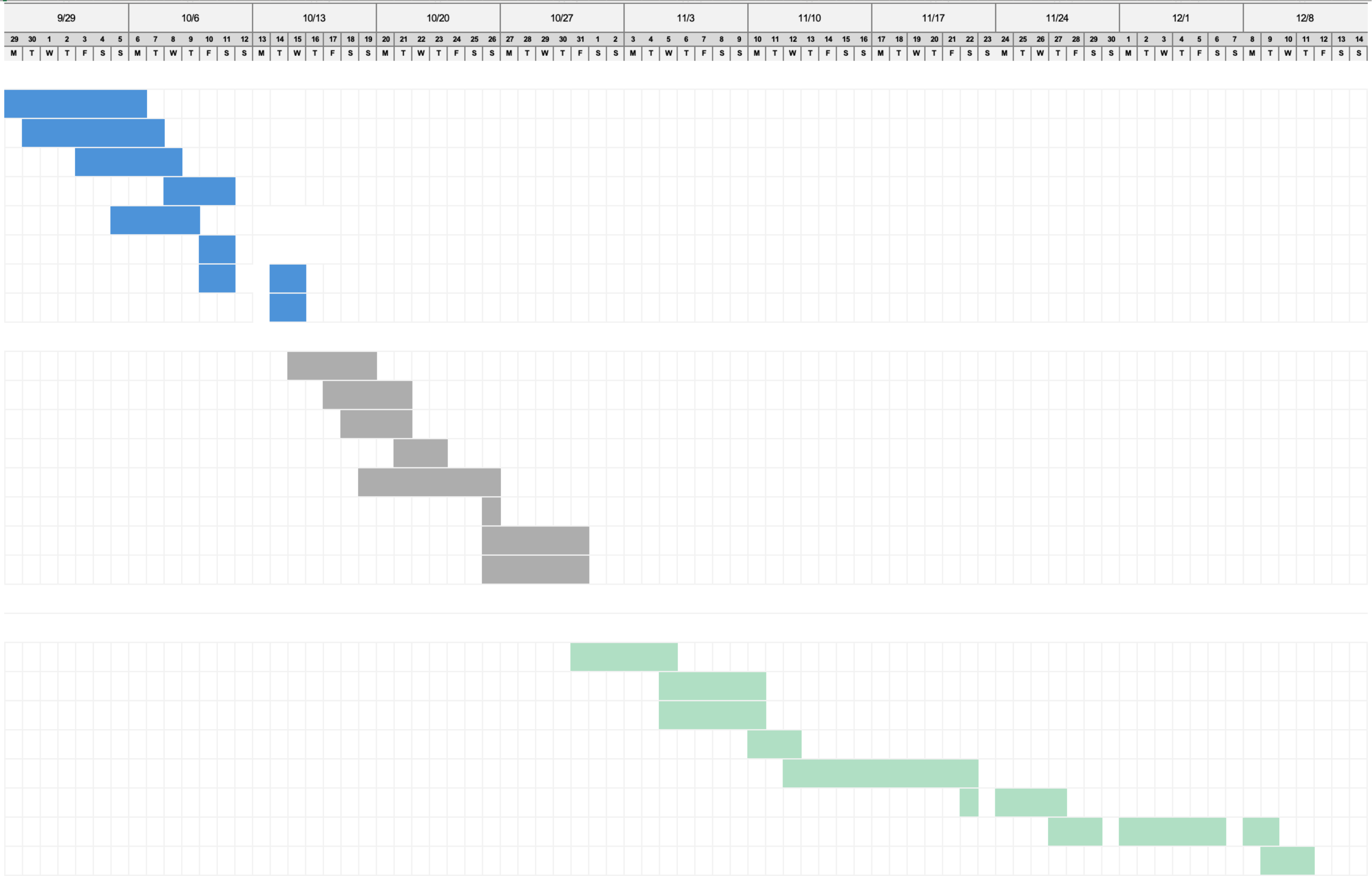
Component	Cycle Time (min)	Quantity
Base	3.78	1
Axle Holder	0.48	4
Gear Block	1.02	1
Motor Mount	0.95	1

Post-Processing: Mill an extra 1mm on interfacing surfaces to ensure flat, smooth, and accurate surfaces for proper assembly and fit.



Project Plan

TASK	START	END
Phase 1: Design & Analysis		
Sketch Drivetrain Design	9/29	10/6
Identify Constraints (Shaft Size, Gear Options, Torque)	9/30	10/7
Preliminary CAD Models	10/3	10/8
Perform Torque and Gear Ratio Analysis	10/8	10/11
Preliminary Materials List (BOM) & Budget	10/5	10/9
Create Manufacturing CAD Model (Adjusting for Extra Tolerances)	10/10	10/12
Finalized Detailed CAD and Dimensioned Drawings	10/10	10/15
Design Review Presentation	10/15	10/15
Prototype Fabrication & Validation		
3D Print finalized DriveTrain subassembly	10/15	10/19
Cut components via Waterjet	10/17	10/21
Drill, Ream, and Tap Critical Holes	10/18	10/21
Test Fit & Functional Prototype Validation	10/21	10/23
Adjust Design Tolerances & Interfacing Issues in CAD	10/19	10/26
Working Prototype of Sub-Assembly	10/26	10/26
All Teams Prototype Car Assembly	10/26	10/31
Developed Preliminary Manufacturing Operation Sheets	10/26	10/31
Phase 3: Batch Manufacturing, Assembly, & Testing		
Design and Machine Assembly Jigs and/or Fixture	10/31	11/2
Finalized Manufacturing Drawings & Operation Sheets	10/31	11/5
Waterjet Cutting of Early Manufacturing Batch	11/5	11/10
Sanding, Machining, Drill/Tap/Ream Axle Holes for Preliminary Batch	11/5	11/10
Manufacturing Review Presentation	11/10	11/12
Finalized Batch Manufacturing	11/12	11/22
Assembly & Integration	11/22	11/27
Fine-Tuning & Full-Vehicle Testing	11/27	12/9
RC Cah Race Event	12/9	12/11



Project Plan

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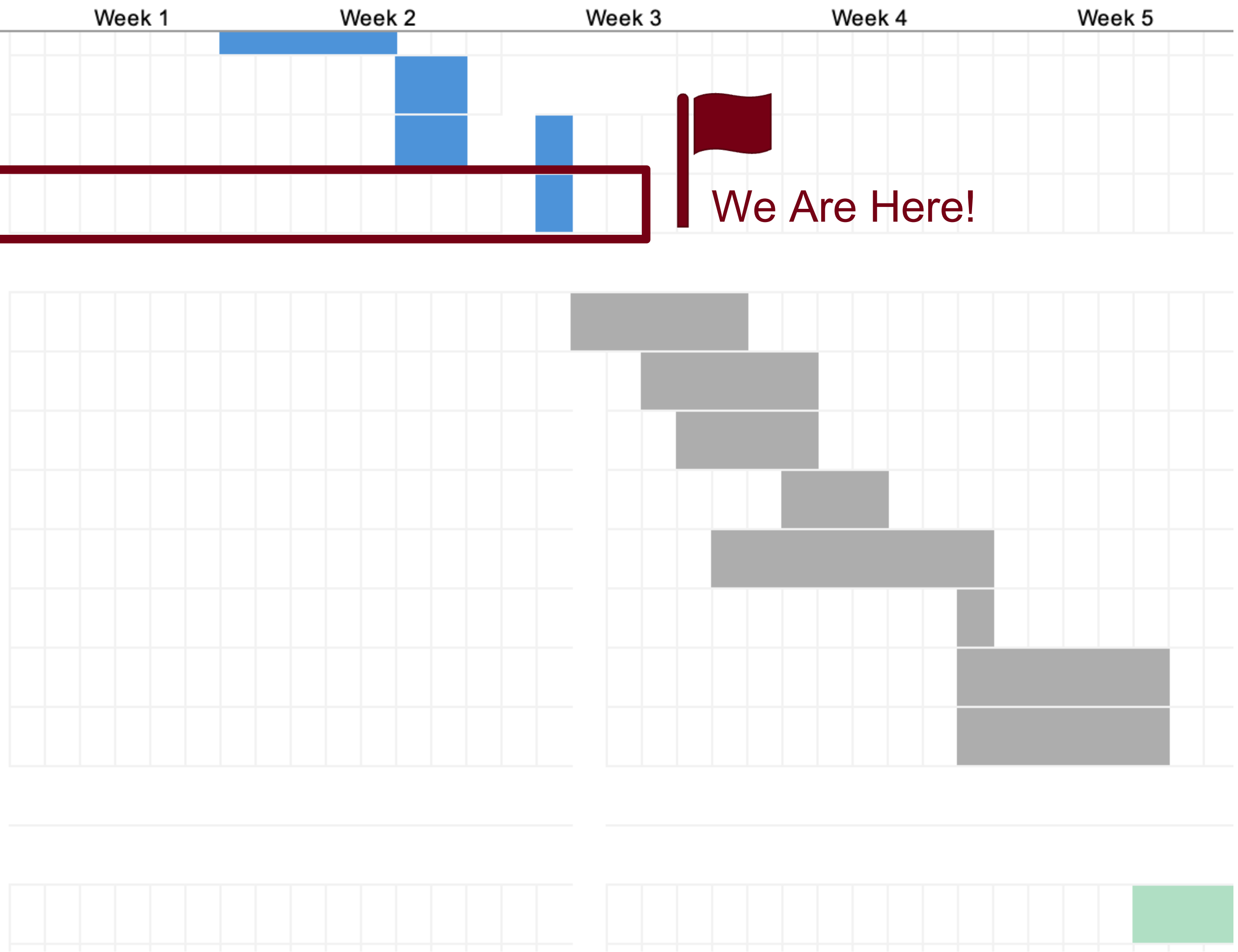
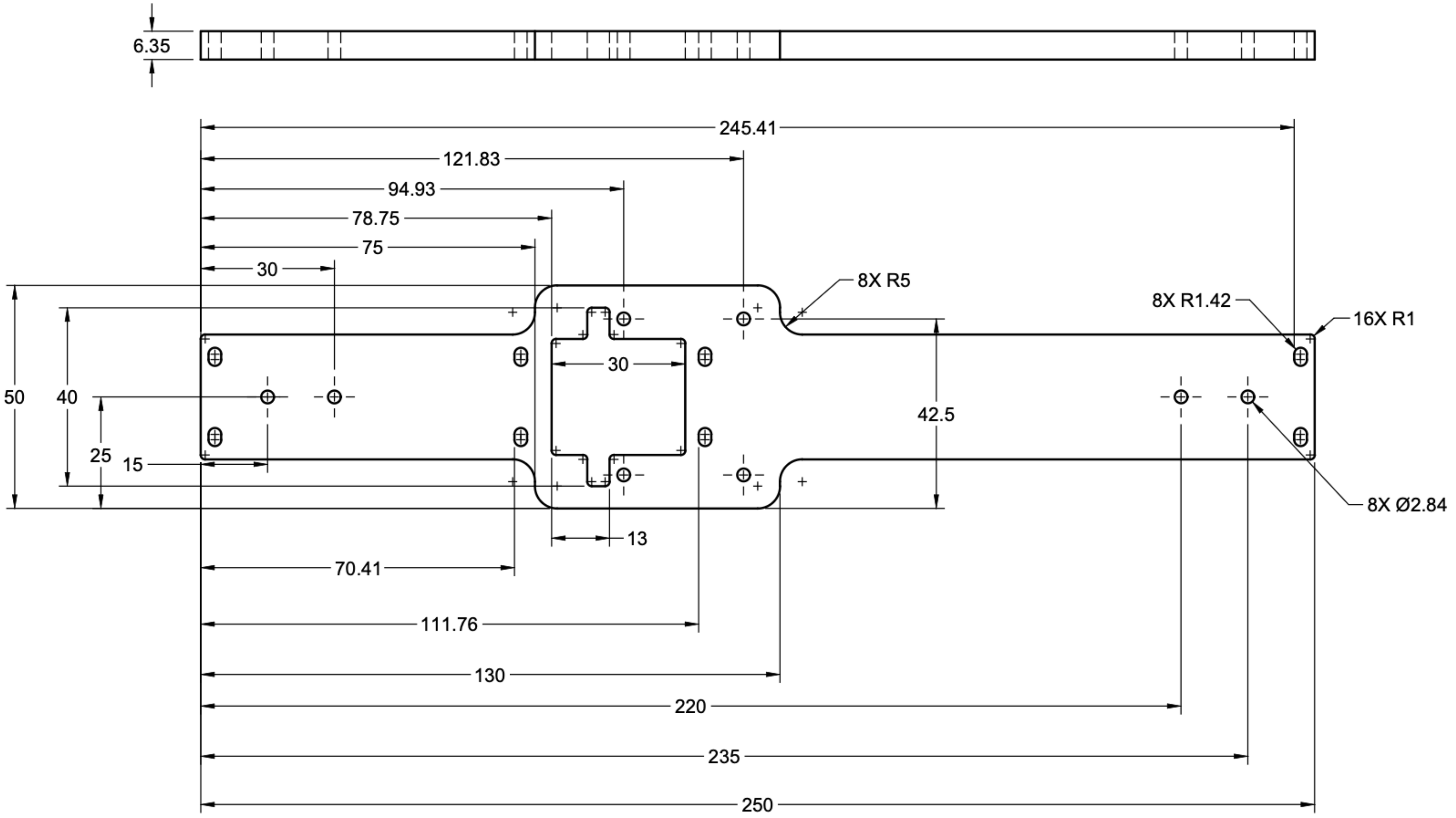


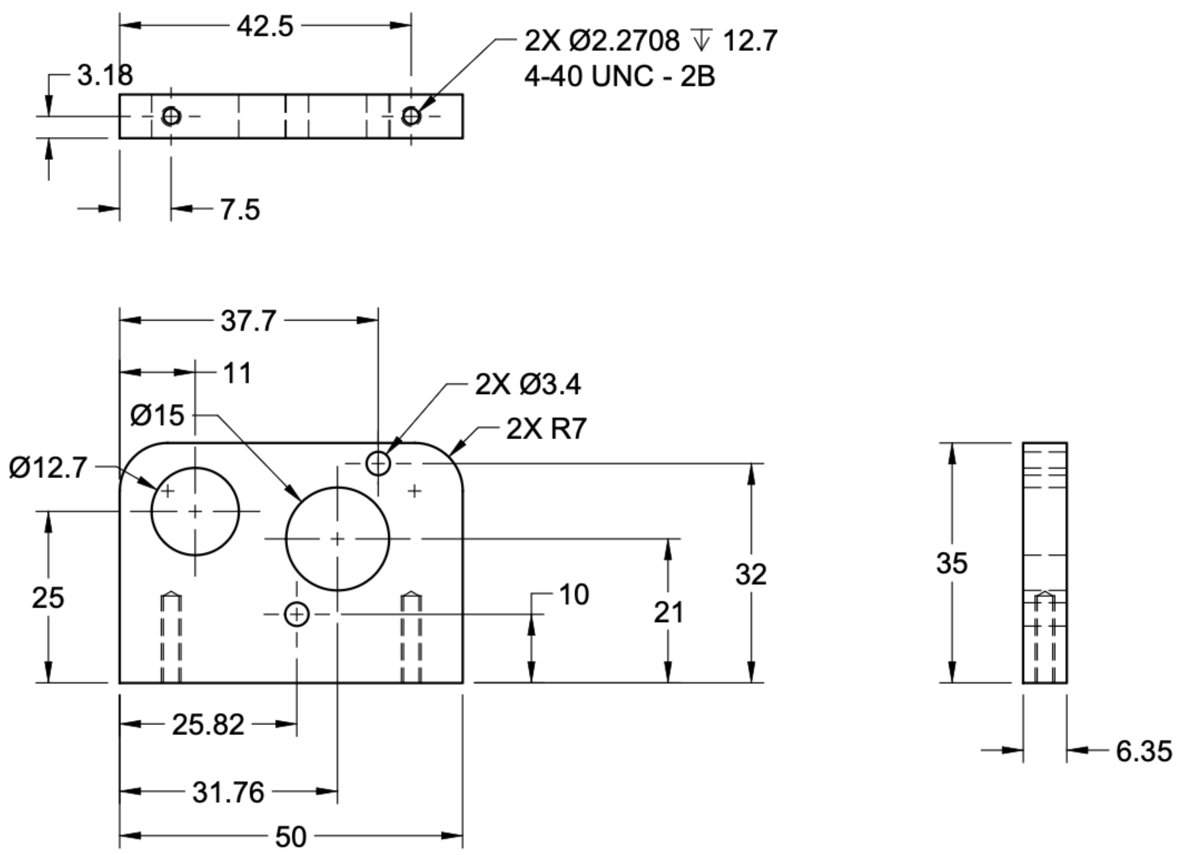
Thank you!

Any Questions?

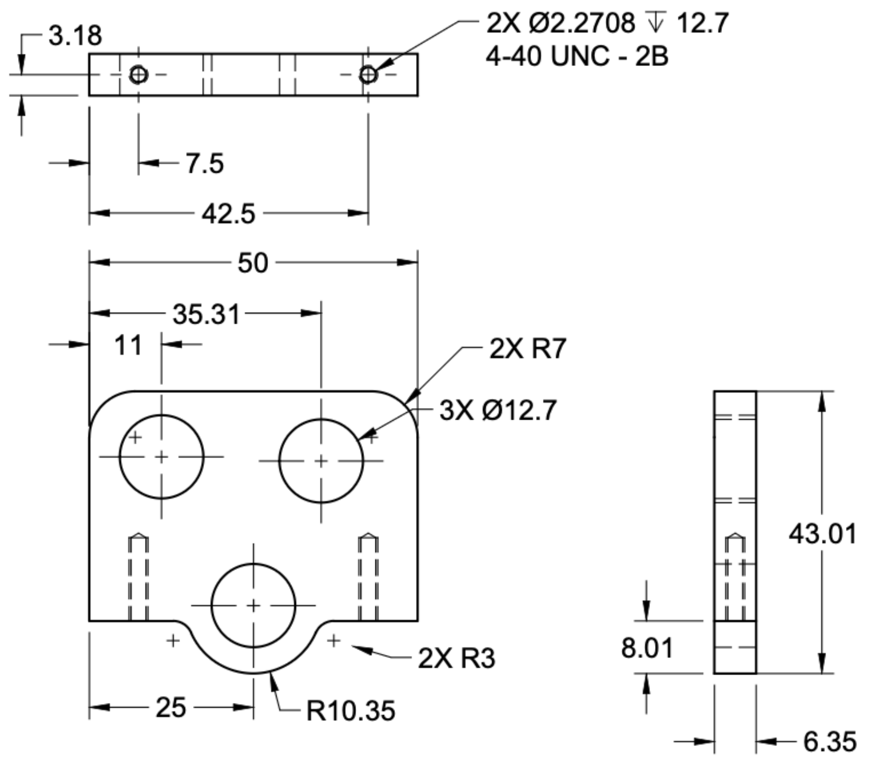
Appendix



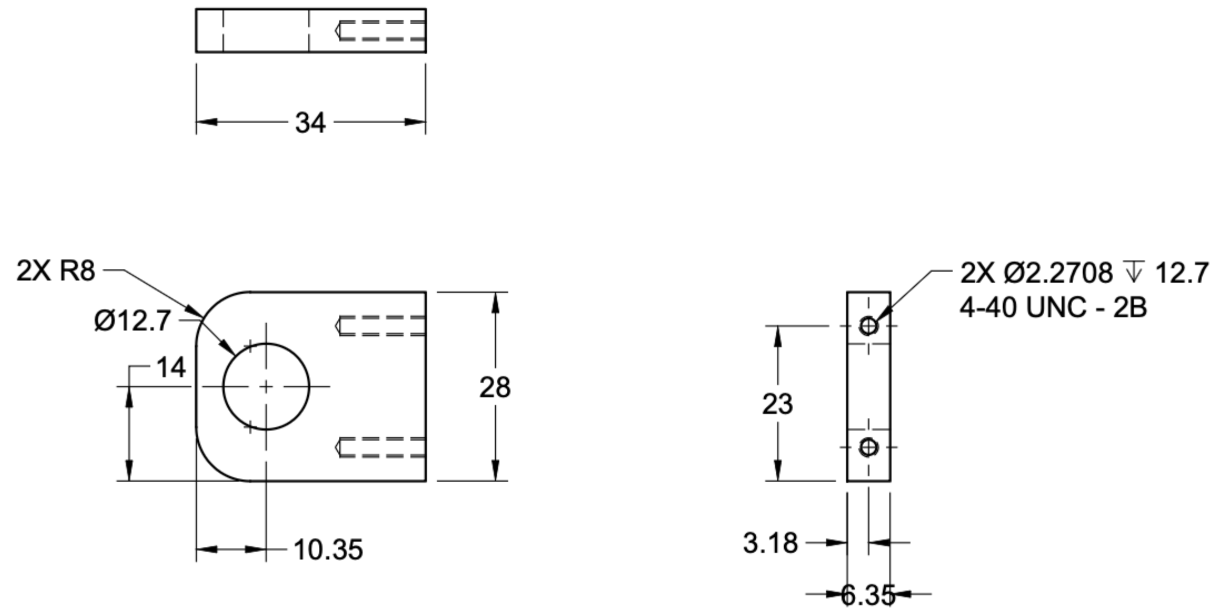
	PROJECT 2.810 FA 2025			
	TITLE Base			
APPROVED	SIZE B	CODE	DWG NO 001	REV 2
CHECKED				
DRAWN Steve Rosario 10/14/25	SCALE 1:1	WEIGHT 123.982 g	SHEET 1/1	



PROJECT	2.810 FA 2025			
TITLE	MotorMount			
APPROVED	SIZE	CODE	DWG NO	REV
CHECKED	A		002	2
DRAWN Steve Rosario 10/14/25	SCALE 1:1	WEIGHT 23.847 g	SHEET 1/1	

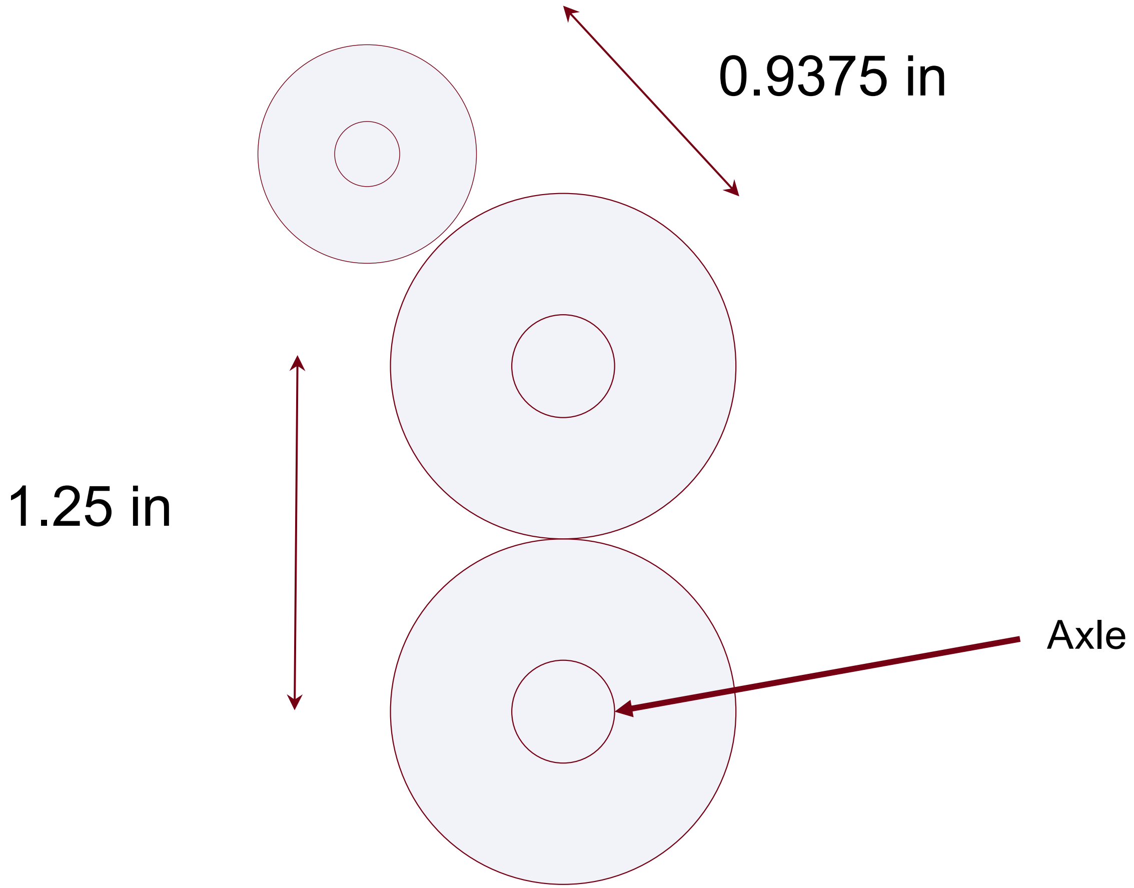


	PROJECT 2.810 FA 2025				
	TITLE GearBlock				
APPROVED	SIZE	CODE	DWG NO		REV
CHECKED	A		003		2
DRAWN Steve Rosario 10/14/25	SCALE 1:1		WEIGHT 24.934 g	SHEET 1/1	



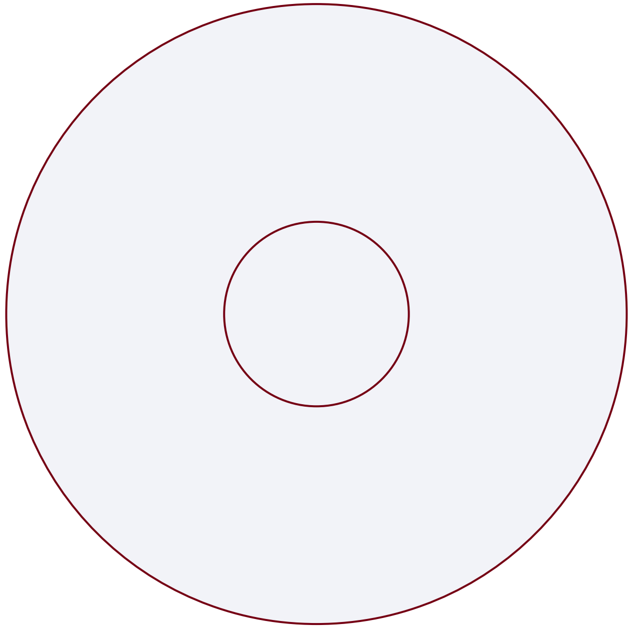
PROJECT				
2.810 FA 2025				
TITLE				
AxleHolder				
APPROVED	SIZE	CODE	DWG NO	REV
CHECKED	A		001	2
DRAWN Steve Rosario 10/14/25	SCALE 1:1	WEIGHT 13.397 g	SHEET 1/1	

Calculations - First Layer

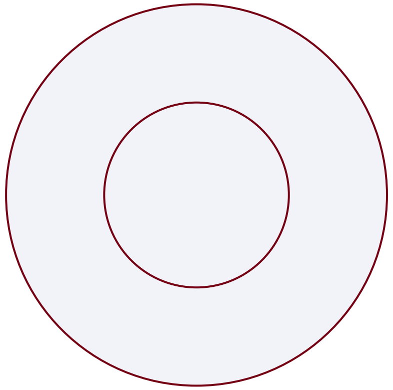


Calculations

<u>40 Tooth gear</u>
Pitch Diameter = 1¼ inch
Inner Diameter = ¼ inch



<u>20 Tooth gear</u>
Pitch Diameter = ⅝ inch
Inner Diameter = ¼ inch



<u>12 Tooth gear</u>
Pitch Diameter = ⅝ inch
Inner Diameter = 0.063 inch